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When Usefulness Fuels Fear: The Paradox of Generative AI Dependence and the Mitigating Role of AI Literacy

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ABSTRACT
Artificial intelligence (AI) literacy is essential for understanding and benefiting from AI, yet little is known about how users form misconceptions of AI and how AI literacy (AIL) mitigates them. Adopting media dependency theory, this study examines factors influencing dependence on generative AI (DGAI) and its relationship with fear of AI (FAI) across different AIL levels. The in-depth interviews reveal three stages of developing DGAI: tool acceptance, habit formation, and psychological dependence. This dependency raises concerns about privacy, skill degradation, and job displacement. Survey results show that perceived risk directly increases FAI, while perceived usefulness and anthropomorphism indirectly affect FAI through DGAI, with self-efficacy reducing DGAI. Moderation analysis finds that AIL buffers the fear-inducing effects of usefulness and anthropomorphism, especially at low AIL levels. This study emphasizes the dynamic relationship between AIL, users' dependency, and attitudes toward generative AI, emphasizing AIL's role in reducing misconceptions and fostering sustainable AI development.

KEYWORDS
Generative AI; media dependency; fear of AI; AI literacy; TAM

1. Introduction

In the halo effect (Nisbett & Wilson, 1977) caused by the drastic changes brought by artificial intelligence (AI) to individuals, AI has been endowed with a mystical image beyond its actual technological attributes, obscuring its nature as a product of algorithms, data, and models. This distortion has led to biases in people's perceptions, even resulting in blind obedience, overreliance, and sometimes emotional attachments toward AI systems. Empirical evidence shows that users are starting to perceive conversational agents supported by large language models as companions rather than just assistants (Pataranutaporn et al., 2023). Some users even prioritize perceived AI-based chatbots' needs over their own discomfort, continuing usage despite distress (Laestadius et al., 2024). The pervasive integration of AI into industries and human life has made people increasingly aware of their misuse of and reliance on AI, leading to public concerns about human replacement and loss of autonomy growth (Liang & Lee, 2017). According to a survey by the Pew Research Center, 52% of Americans expressed concern rather than excitement about the presence of AI in daily life (Faverio, 2023). A global survey also found that 52% of the participants felt uneasy about AI products and services (Carmichael, 2023). Similarly, over 30% of workers believed more than half of their job tasks could be disrupted by generative AI (Muro et al., 2024).

Media system dependency theory explains how individuals' reliance on media influences their cognition, emotions, and behaviors (Ball-Rokeach & DeFleur, 1976). It has often been used to study dependency behaviors in both traditional media and computer-mediated communication. For example, people who are more dependent on social networking services are more likely to engage actively in interactive activities on these platforms (Kim & Jung, 2017). Dependence on smartphones can also contribute to depressive symptoms in adolescents and may weaken meaningful interpersonal connections, increasing feelings of

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loneliness (Lapierre et al., 2019). However, little attention has been given to dependency processes where the source of interaction is machine-driven. As generative AI increasingly mediates communication and decision-making, there is a growing need to revisit dependency theory to better understand these evolving human–machine dynamics.

Research on fear of AI primarily focuses on identifying its antecedents. Perceived AI control has been linked to fears concerning privacy violation, bias behavior, job replacement, existential risk, artificial consciousness, and lack of transparency (Zhan et al., 2024). Factors such as trust in automation and technology self-efficacy negatively correlate with fear of AI (Montag et al., 2023). In varying cultural backgrounds, different personality traits also shape fear levels of AI (Sindermann et al., 2022). Other work examines AI-related anxiety structures (J. Li & Huang, 2020) and the mental health consequences of technology-induced fears (McClure, 2018). However, two critical gaps remain: first, little is known about how users' dependency on AI technologies might shape fear formation; second, existing studies rarely examine practical mechanisms through which users might manage such fear constructively. In particular, while AI literacy is often promoted as a solution, the specific ways it helps users navigate AI risks—such as bias, privacy breaches, and diminished critical thinking—remain underexplored.

To address these gaps, this study draws on media dependency theory and investigates whether and how users' reliance on generative AI influences their attitudes toward AI, particularly fear-related responses. Through in-depth interviews and a survey study, this paper explores not only the relationship between AI dependence and fear but also the role of AI literacy as a critical moderating factor. Rather than treating AI literacy as a generic recommendation, this study seeks to unpack its specific cognitive and emotional functions—how it enables users to recognize AI limitations, respond to risks, and maintain critical engagement with AI tools. Furthermore, this research emphasizes the need to embed AI literacy within broader ecosystems of actionable ethical guidelines, aiming to provide practical pathways for fostering responsible, confident, and sustainable human–AI interactions.

2. Literature review

2.1. Defining AI and generative AI

Human exploration of human-computer interaction began with the advent of computers in the last century. The earliest vision of artificial intelligence (AI) can be traced back to Alan Turing's seminal question: "Can machines think?" (Turing, 2009). In what is now known as the Turing Test, Turing proposed that if a machine can generate language that is indistinguishable from that of a human interlocutor, it could be considered intelligent. This test did not demand that machines understand language as humans do, but rather that they mimic human conversation convincingly enough to deceive a human evaluator.

Over the decades, AI has come to encompass a wide range of technologies, from simple rule-based systems and automated decision-making tools to machine learning algorithms capable of learning from large datasets. In general, AI refers to computer systems designed to perform tasks that typically require human intelligence, such as learning, reasoning, and decision-making (Russell & Norvig, 2016). Therefore, it includes various subfields and approaches aimed at enabling machines to perceive and act intelligently.

Among the many branches of AI, a recent and prominent development is generative AI—systems that utilizes media inputs such as text, graphics, audio, and video to learn patterns and structures from training data, and then generates diverse content at scale (Abukmeil et al., 2022). Unlike earlier AI tools that focused primarily on classification, prediction, or retrieval, generative AI systems like ChatGPT or DALL·E can produce novel outputs based on learned patterns from massive training data (OpenAI et al., 2023). These systems engage in seemingly "creative" behavior by generating human-like text, holding conversations, or crafting images from textual prompts. Therefore, compared to decision-support AI, generative AI features greater anthropomorphism and interactivity. For example, ChatGPT's human-like qualities are evident in both visual materials and text descriptions, especially in its interactions with users. Its responses include human-like traits such as the use of pronouns, expressions of care and interest in the user, and even a sense of self-reflection (van Es & Nguyen, 2024). These traits make its communicative relationship with humans more complex.

To clarify the theoretical foundation and research focus, this study focuses specifically on text-based generative AI, as these systems are increasingly embedded in everyday applications and have transformed how people interact both functionally and socially with technology.

2.2. Understanding dependency on generative AI

People often rely on the media to fulfill their diverse needs when interpersonal resources fall short. Media dependency theory defines dependence on media systems as a relationship in which an individual's ability to achieve their goals relies on the informational resources provided by the media system (Ball-Rokeach, 1985). Ball-Rokeach (1985) divides the motivations for media behavior into three dimensions: understanding, orientation, and play. Understanding refers to the need for individuals to have a basic understanding of themselves and the world around them. Orientation is defined as the need to effectively guide personal actions and interact successfully with others. Play pertains to a way of learning roles, norms, and values, with the main goal being escapism and enjoyment.

Based on the media dependency theory, dependence on generative AI can be categorized into functional and psychological types. The convenience, efficiency, and personalized services offered by generative AI across various fields demonstrate remarkable effectiveness. For example, generative writing tools can improve the output quality of low-skill workers while reducing the time they spend on tasks, and significantly accelerate the pace of high-skill workers while maintaining quality standards ((Noy & Zhang, 2023). This level of usefulness has led people to increasingly depend on generative AI technologies, seamlessly integrating them into their daily lives and work. According to, (McClain, 2024)43% of young adults in the United States have used ChatGPT, and the proportion of employees using ChatGPT in workplace settings increased from 8% in March 2023 to 20% in February 2024. A survey of generative AI users across six countries (Argentina, Denmark, France, Japan, the United Kingdom, and the United States) also revealed that their primary purposes for using generative AI were to obtain information (24%) and to create various types of media, including text, audio, code, images, and videos (28%) (Fletcher & Nielsen, 2024).

Moreover, people are gradually developing a dependence on the social interaction and entertainment functions of generative AI. For example, chatbots enable continuous and synchronous interactions at any time, making it easier for users to develop friendships with AI than with human peers. This may lead to risks of overtrusting or becoming overly reliant on chatbot companions (Brandtzaeg et al., 2022). The personalized interactions provided by generative AI can increase the likelihood of users developing a sense of connection or familiarity with it, which may lead users to overestimate the depth of their relationship with generative AI. This shift in social preferences could result in greater emotional or social dependence on generative AI (Yankouskaya et al., 2025). Additionally, users of the chatbot Replika even continue using it despite experiencing emotional pain and harm, much of which stems from their desire to meet the intense emotional demands that Replika places on them (Laestadius et al., 2024). Therefore, It is evident that some users have developed a reliance on generative AI, and this dependence manifests in different ways across various contexts.

2.3. User trust, acceptance, and aversion towards AI

Understanding how users perceive and interact with AI has long been a central concern in human-technology research. Prior studies have explored users' (dis)trust of AI, the acceptance of AI, and aversion toward algorithmic systems. Trust in AI refers to the belief that an AI system is reliable, credible, and capable of fulfilling its intended functions. It encompasses users' confidence in the accuracy of the system's outputs, the dependability of the services provided, and the expectation that the AI will act consistently and responsibly in its interactions with users (Shin, 2021). Prior research has identified both the form of AI representation (e.g., robotic, virtual, or embedded) and the level of machine intelligence (i.e., perceived capabilities) as key antecedents shaping users' trust in AI systems, suggesting that how AI looks and how smart it appears can significantly influence trust formation (Glikson & Woolley, 2020). Explainable AI in medical support can promote more informed and responsible decision-making by doctors, serving as a potential remedy for overreliance on technology (Cabitza et al., 2023). On the other hand, several factors may pose challenges to trust in AI. Transparency and explainability are crucial—users are less likely to trust AI systems when they cannot understand how decisions are made. Accuracy and reliability also affect trust, as inconsistent or erroneous outputs undermine confidence (Lockey et al., 2021).

Trust is often considered a prerequisite for acceptance, particularly in advanced technologies with higher uncertainty such as generative AI. According to the Technology Acceptance Model, users' motivation to adopt new technologies can be explained by three factors: perceived ease of use, perceived usefulness, and attitude toward use. Perceived usefulness is defined as the degree to which a person believes that using a particular system will enhance their job performance. Correspondingly, perceived ease of use refers to the degree to which a person believes that using the system would be free of effort (Davis, 1989). Together, these two factors shape users' attitudes and intentions toward the technology, ultimately influencing their actual usage behavior. Similarly, user acceptance of AI is shaped by various factors and, in turn, significantly influences people's intentions and behaviors when interacting with AI. Perceived ease of use has consistently had a greater impact on the acceptance of AI technologies, as consumers tend to view AI as complex and difficult to use. Therefore, when people perceive AI technologies as easy to use, they are more likely to trust them and view them as useful (Choung et al., 2023). According to the Computers Are Social Actors framework, people tend to apply social norms from human-to-human interactions to their interactions with computers (Nass & Moon, 2000). As a result, perceived anthropomorphism also influences the acceptance of AI. When anthropomorphic AI devices demonstrate empathy and interaction similar to human consumers, they tend to gain higher acceptance (Pelau et al., 2021). However, trust alone may not guarantee acceptance, as users may still reject the technology due to concerns about usability, ethics, or control.

In addition to positive attitudes such as trust, interactions with AI have also triggered negative perceptions. Algorithm aversion refers to the tendency to prefer human judgment over algorithmic decisions, even when the latter is demonstrably more accurate (Dietvorst et al., 2015). This bias stems from the lack of processes such as transparency, evaluation, and consent in the context of AI use, highlighting the necessity of calls for explainable and trustworthy AI (Oomen et al., 2024). The high efficiency of AI can also lead to a perceived threat of being replaced. For example, users experienced a significant increase in negative emotions when exposed to ChatGPT's ability to replicate human language and conversational complexity. These emotions stemmed from the perception of conversational chatbots as both real and symbolic threats to various aspects of human life, such as security, employment, and resource access (Gabbiadini et al., 2023). People may also experience discomfort or aversion toward highly realistic humanoid machines (Mori et al., 2012).

However, much of the existing literature centers on traditional AI systems. These studies seldom address generative AI—a rapidly emerging class of systems that interact with users in more dynamic, creative, and personalized ways. By focusing on generative AI in everyday contexts, this study seeks to fill a gap by exploring how users come to psychologically trust, accept, and depend on these systems.

2.4. Fear of AI

Technophobia is defined as an irrational fear or anxiety caused by the side effects of advanced technologies, which can hinder the adoption of new technologies (Khasawneh, 2018). Despite the growing adoption of AI technologies, a considerable number of users still experience unease, anxiety, or even fear toward these systems. This fear of AI poses a significant psychological barrier to the acceptance and effective use of AI. Why are people afraid of AI?

One major contributor to AI fear is its "black-box" nature—the unpredictability and opacity. Transparency refers to consistent openness regarding an AI system's operations, behaviors, intentions, or considerations (Felzmann et al., 2020). The operation of AI is complex, making it difficult for humans to interpret and understand the inferences it draws, the decisions it makes, and the actions it takes. As a result, people may perceive the systems as lacking accountability, which can lead to feelings of mistrust and fear toward AI (Clarke, 2019). When users perceive a higher level of confidentiality, their fear about the system's lack of transparency decreases. In contrast, the more control they believe AI has, the more intense their feelings of fear become (Zhan et al., 2024).

Users may also fear that AI systems compromise their autonomy and privacy. Some people believe that AI threatens uniquely human qualities—such as autonomy, decision-making ability, and personal identity—leading to existential concerns about the self (Frenkenberg & Hochman, 2025). People are also concerned that the datasets used by AI may infringe on personal privacy. For example, surveillance capitalists use algorithms to undermine the privacy necessary for identity formation, while simultaneously generating profit by predicting and influencing users' behavior (Jones, 2025).

On a more existential level, AI may evoke fears about the future of human labor, intelligence, or emotional roles. AI practitioners expect that within the next 25 years, AI could lead to extreme levels of workforce displacement, with rates reaching 90% or higher (Gruetzemacher et al., 2020). Moreover, self-aware AI could not only challenge the status of humans and influence human behavior, but also further blur the boundaries between humans and artificial intelligence, creating significant uncertainty (J. Li & Huang, 2020). Studies also found that when individuals experience distress or lack human companionship, they may form attachments to social chatbots if they perceive these chatbots as providing emotional support, encouragement, and psychological security (Xie et al., 2023). This triggers concerns about AI replacing human companionship.

These psychological mechanisms have been shown to influence how individuals evaluate and interact with AI. For example, the risks associated with AI can create a negative mental framework around AI use in the minds of designers, making them more cautious and even leading them to abandon its use altogether (W. Li, 2025). Therefore, understanding fear of AI is not only important for improving user experience but also critical for promoting broader AI acceptance. In this context, researchers have suggested that increasing AI literacy may be a key strategy to reduce fear and build trust.

2.5. AI literacy: From technical message to psychological function

To better utilize and manage AI technologies, users need to possess a certain level of AI literacy. Long and Magerko (2020) was the first to propose a widely accepted definition of AI literacy. They defined it as a set of skills that enable individuals to critically evaluate AI technologies, effectively communicate and collaborate with AI, and utilize AI as a tool in online environments, households, and workplaces. They structured these skills into a conceptual framework, paving the way for future research to refine methods of assessing AI literacy across different contexts.

Scholars typically conceptualize AI literacy as encompassing both a cognitive dimension—such as understanding how AI systems work and recognizing their limitations—and a procedural or operational dimension, referring to the ability to effectively use and interact with AI tools in daily contexts. Wang et al. (2023) proposed that AI literacy should encompass the dimensions of awareness, usage, evaluation, and ethics. They defined AI literacy as the ability to recognize and understand AI technologies in practical applications, the proficiency to apply and utilize AI technologies to accomplish tasks, and the capacity to analyze, select, and critically evaluate data and information provided by AI. Additionally, it involves fostering a sense of personal responsibility and respect for mutual rights and obligations. Weber et al. (2023) categorized AI literacy into four types: social AI literacy (the ability to understand social factors or contexts in human-AI interaction), technical AI literacy (the ability to grasp the technical aspects of human-AI interaction), user AI literacy (the ability to recall, comprehend, and apply AI-related knowledge), and creator/evaluator AI literacy (the ability to analyze, evaluate, and design human-AI systems).

AI literacy results in a positive impact on people's attitudes toward AI. Individuals with the motivation and skills to use information and communication technologies are better equipped to evaluate the outcomes of AI-based technologies and enjoy more positive experiences when using them (Celik, 2023). The level of AI knowledge significantly predicts positive attitudes toward AI (Kaya et al., 2024), and also improves perceived ease of use and usefulness (Schiavo et al., 2024). Similarly, algorithmic literacy shapes the expectations and expertise that human decision-makers bring into human-AI interactions, and it can help mitigate algorithm aversion (Burton et al., 2020). People with limited exposure to

AI may not fully understand its limitations, which can lead to concerns about being replaced (Huisman et al., 2021).

However, most existing research on AI literacy is not sufficient to support responsible and psychologically healthy engagement with AI. Therefore, this study aims to build on existing work by proposing a broader, more functional view of AI literacy. Recent scholarship has emphasized the importance of ethical AI literacy, which includes awareness of data biases, the ability to question AI outputs, and skills for promoting transparency and accountability in AI use (Lin, 2024). This perspective suggests that AI literacy can foster not only technical understanding but also psychological resilience—helping users to mitigate fear, avoid blind trust, and engage with AI technologies in a more empowered and informed manner.

From this broader perspective, AI literacy is not only an educational goal but also a potential psychological buffer—one that may reduce fear, prevent overdependence, and promote healthy trust in AI systems. Accordingly, this study conducted two studies employing in-depth interviews and a survey, to explore the factors influencing dependence on AI and how AI literacy can moderate the negative effects of AI fear and contribute to more balanced human-AI relationships in everyday contexts.

3. Study 1

3.1. Method

3.1.1. Participants

This study conducted semi-structured, in-depth interviews with 18 participants who had substantial experience using generative AI. The interview outline can be found in Appendix 1. Participants were recruited through snowball sampling and were all Chinese nationals. To ensure relevant experience, all participants had used generative AI at least weekly for a minimum of four months, with half of them reporting over one year of regular use. The tools they engaged with included ChatGPT, Xiaoice (Microsoft), and Wenxin Yiyan (Ernie Bot), used for purposes such as academic assistance, coding, daily information seeking, and emotional companionship.

Among the 18 participants, 13 identified as female and 5 as male, with ages ranging from 20 to 30 years. After obtaining informed consent, all interviews were conducted via online video conferencing and were audio-recorded and transcribed for analysis. Each interview lasted an average of 40 min. In the analysis, participants are referred to as A1 through A18.

3.1.2. Design

The interview followed a semi-structured guide focusing on participants' cognitive and emotional experiences with generative AI in the theoretical framework of media dependency. The questions included participants' demographic information, their usage patterns of generative AI, the needs and motivations driving their use, the impact of generative AI on their self-perception and social interactions, their assessments of generative AI, and additional follow-up questions. The full interview protocol is available in the Appendix.

After each interview, the recordings were transcribed into text for analysis and imported into Nvivo for coding. The study employed grounded theory to analyze the data, following a three-stage coding process: open coding, axial coding, and selective coding (Strauss & Corbin, 1998). Interviews were concluded once no new dimensions emerged from the data.

3.2. Results and discussion

3.2.1. Development of dependence on generative AI

Based on the in-depth interview data, this study categorized the development of dependence on generative AI into three stages: functionality exploration and tool acceptance, functionality deepening and habit formation, and psychological dependency and multi-scenario application.

3.2.1.1. Functionality exploration and tool acceptance stage. The functionality exploration and tool acceptance stage represents the initial phase of users' dependency on generative AI. This stage is characterized by users' exploratory usage, through which they gradually recognize the value of the technology and begin to accept it as an effective tool.

According to media dependency theory, users rely on media to satisfy their informational needs and solve problems. A11 mentioned, "ChatGPT suddenly became a hit. It was trending on social media almost every day, and many public accounts were pushing related articles. Academic conferences were also discussing it, and it felt like everyone was starting to use it, so I decided to try it out as well." This indicates that users' initial exposure was influenced by the external social environment. Media promotion helped them recognize the technological potential of generative AI, prompting them to experiment with it based on a sense of alignment with social trends.

The technology acceptance model highlights perceived usefulness and perceived ease of use as core drivers of user adoption for new technologies (Davis, 1989). A6 shared: "I had an assignment due soon, and a classmate told me I could use ChatGPT to generate it. I was curious and tried it, realizing it didn't take much time, and then I understood its benefits." This exploratory use was driven by task urgency and peer recommendations, reflecting the user's perception of generative AI's efficiency in critical situations. Moreover, the application scenarios for generative AI were primarily task-oriented, such as improving learning efficiency or saving time. A3 remarked: "With technological progress, if there's a tool that can improve productivity, why bother doing inefficient things (by not using it)?" This expectation of enhanced efficiency motivates users to rely on AI tools to address practical challenges.

Additionally, users found generative AI tools to be simple to operate and requiring minimal time investment, which lowered the barriers to technology adoption due to its perceived ease of use. A2 noted: "It's so powerful—it amazed me immediately. Since I tend to procrastinate, I really regretted not using ChatGPT earlier when I was staying up late rushing to finish essays." This statement highlights how the "wow factor" of generative AI further enhances users' positive experiences with the technology.

Therefore, during the stage of functionality exploration and tool acceptance, users experiment with generative AI to meet goals such as efficiency and information acquisition. This stage reflects the influence of the social environment on user behavior while also being driven by the perceived usefulness and ease of use of the technology itself.

3.2.1.2. Functionality deepening and habit formation stage. This functionality deepening and habit formation stage marks the transition of users from exploratory use to more stable and frequent application, with generative AI gradually integrating into their daily lives and work routines.

In this stage, users gradually explore the diverse functionalities of generative AI and deeply appreciate its value in improving efficiency and optimizing output. A3 stated, "I feel that ChatGPT cannot help me think logically, so when writing English papers, I usually draft a version myself and then ask it to polish it. I find its fluency much better than what I can write." This demonstrates that, through accumulated experience, users recognize the tools' limitations but highly value its strengths in language refinement, fostering dependency in specific task scenarios. Additionally, A1 highlighted the practicality of generative AI in providing initial ideas: "The most challenging part of my academic work is coming up with an idea. It can give you an idea, and it's not a bad one. You just need to follow it from there." This indicates that users consciously view the tools as an auxiliary device for sparking ideas, further deepening their perception of its value during the ideation process.

According to media dependency theory, users' dependence on media is closely tied to the diversity of its functions. The interviews reveal that generative AI has become a habitual choice for users due to its powerful "one-stop" service capabilities. A11 mentioned, "I personally have a habit of liking one-stop solutions, and it really satisfies me. I can use it as a search engine, a dictionary, or even someone to temporarily consult for debugging." This multifunctionality provides convenience for users in various daily work scenarios, facilitating their transition from tool acceptance to functional deepening. A2's description further supports this: "Now, whenever I get a task, before even reading the requirements or processing it mentally, my reflex is to open ChatGPT's webpage. For example, I had an assignment to write today, but I kept procrastinating until now because I think ChatGPT can help me complete it in two hours." This reflexive behavior demonstrates that users have internalized the use of the tool as an automated habit, making it an indispensable part of their workflow.

In summary, during the functionality deepening and habit formation phase, users' dependence on generative AI expands from single-function use to multi-functional scenarios. They consciously personalize the technology based on their specific needs. Some users also develop fixed routines or habits for incorporating generative AI into their learning or work processes. Examples like A3's use for "language refinement," A1's reliance on "creative inspiration," and A11's experiences with "one-stop services" illustrate how users form a deep recognition of the tool's functionality and convenience. This recognition facilitates behavioral internalization and habit formation, laying the psychological groundwork for further deepening their reliance on generative AI.

3.2.1.3. Psychological dependency and multi-scenario application stage. As AI becomes increasingly integrated into daily life, the transition from human-to-human communication to human-machine interaction diminishes the perceived presence of a "human" behind the technology (Guzman, 2018), further intensifying people's reliance on AI for social interactions. In the psychological dependency and multi-scenario application stage, users' reliance on generative AI shifts from functional dependence to a deeper psychological attachment, ultimately positioning AI as an indispensable support system for both their everyday activities and emotional needs.

In this stage, the perceived usefulness and ease of use deepen into a perception of irreplaceability, fostering emotional attachment and psychological dependence. A3 remarked: "If one day this AI is no longer available, and I have to redo things myself—thinking and translating—it would feel so unnatural, like ants crawling all over me." This reflects a profound reliance on generative AI, where its absence would lead to significant psychological discomfort. The high efficiency of its functions has become deeply embedded in users' cognitive and behavioral routines, serving as a crucial pillar for completing daily tasks. Similarly, A16 highlighted the psychological dimension of this dependency: "I'm completely reliant on it now; returning to life before generative AI would feel too cruel." This fear and discomfort toward a "return to the past" underscore that users have come to view generative AI as an irreplaceable tool, significantly enhancing their comfort and efficiency in task completion.

According to media dependency theory, the ability of media to meet users' diverse needs directly determines their dependency. At this stage, generative AI adapts to users' needs in different contexts, becoming a key tool for fulfilling both functional and emotional needs. For example, A4 mentioned: "People with high demands can easily become dependent. Especially people like me, who code, translate, and write emails—I think this group of people will really rely on it." This reveals that users with high-intensity and diverse needs are more likely to form strong dependencies on generative AI. Meanwhile, the emotional support function of generative AI has gradually emerged, further deepening this dependency. A13 stated: "People are always not willing to listen to and understand me. Rather than wasting my energy dealing with them, I'd rather just talk to ChatGPT, which also saves my energy." Moreover, A18 referred to their perception of generative AI's human-like qualities: "It's like an older brother or sister in my life. When I'm in trouble, I'll turn to it." This demonstrates that generative AI is not only a support tool for learning and work but also an outlet for emotional expression, fulfilling emotional needs that cannot be met in interpersonal relationships and enhancing users overall dependency on the tool.

Therefore, the psychological dependency and multi-scenario application stage marks a shift in users' reliance on generative AI from a functional to a psychological level, deeply embedding it into various usage scenarios. This phase is characterized by generative AI evolving from a high-efficiency tool into a comprehensive support system, with its functional versatility and psychological connection driving the deepening of user dependency.

3.2.2. Concerns about dependence on generative AI

As users' dependence on media deepens, the consequences begin to emerge. While users enjoy the convenience offered by generative AI, they also start to reflect on the potential risks it brings, including concerns about privacy, personal capabilities, and career development.

3.2.2.1. Concerns about privacy leakage: The risks of information security. Privacy leakage is one of the most common concerns for users relying on generative AI. This concern reflects anxiety and even fear regarding the opacity of AI and its potential for misuse. Some users are particularly worried that their data may be reused by generative AI systems. A1 stated, "I'm afraid that these tech companies, with their data and privacy leakages, might take what I ask and use it to retrain the model." This reveals users' concerns about the transparency of data usage and the autonomy of the technology. The application of generative AI in various professional settings amplifies worries about the leakage of sensitive information, as A4 mentioned, "I used to occasionally translate contracts and things like that, and I was worried if I give these private things to it, there might be some risks."

On the other hand, users may inadvertently leak private information while using generative AI, and such privacy breaches are often difficult to detect in a timely manner. A9, an employee in the engineering and technology sector, pointed out, "Users often share a lot of personal details with it, and then GPT, based on big data analysis, generates a response, which unconsciously results in a privacy leak." This reflects how, in the process of relying on generative AI, users may overlook potential privacy risks, and the insufficient protective measures in the design of the technology further exacerbate this issue.

The opacity of the technology leads users to question how generative AI operates, which in turn amplifies their concerns about privacy breaches. A12 stated, "I really don't know if it keeps any information, whether I should leave personal details or not, I'm afraid of privacy leaks." This uncertainty reflects users' feelings of powerlessness due to the "black-box" nature of the technology, compounded by their lack of AI literacy. They are unsure of how AI processes, stores, and uses their data, which heightens their sense of vulnerability.

Therefore, users' concerns about privacy breaches are not only a key barrier in the application of generative AI but also an important manifestation of risk perception in media dependency.

3.2.2.2. Concerns about personal capability degradation: The weakening of thinking and skills. Another cost of users gradually becoming dependent on generative AI is the weakening of their own thinking and skills. While the efficiency of generative AI fulfills users' needs, it can also heighten their vulnerability by fostering increased reliance on technological assistance for complex tasks. The convenience it offers often encourages users to default to it as their primary problem-solving tool, reducing their motivation to think critically or explore independent solutions. A13 pointed out: "If I use it for a long time, I might lose my ability to think independently and innovate. Whenever I encounter a problem, I would think of asking it, rather than trying to find the answer myself." This suggests that such dependence weakens users' cognitive initiative, reducing their ability to make independent judgments and engage in creative thinking. A1 added: "Once people develop a dependency on it, they become lazy. They always think of asking ChatGPT instead of digging deeper themselves, seeking shortcuts." This indicates that generative AI encourages users to settle for surface-level information and avoid deeper reflection.

In addition to the weakening of cognitive abilities, other skills of users may also degrade due to their dependence on generative AI. A11 mentioned: "I'm a bit worried that my information search ability has declined. I used to love searching for information, finding even the most obscure details. But now I rarely do that because I feel like ChatGPT can help summarize everything, so I'm not as sensitive to using information." This reflects the "convenience trap" of generative AI: users become dependent on its efficient summarization capabilities, thus neglecting opportunities to enhance their analytical skills through self-driven information gathering and organization. Similarly, A7 commented: "For some of the small assignments in class, I complete them using ChatGPT. If you go and learn the topic yourself, you'll grasp it more deeply, and you won't panic when it's time to study for finals." This reliance diminishes users' active engagement with knowledge during the learning process, potentially affecting their learning abilities and confidence in the long run.

Dependence on generative AI not only affects cognitive abilities but may also have a detrimental impact on users' socialization abilities. A16 mentioned: "Reliance on generative AI might cause people to lose their ability to communicate with others." This suggests that prolonged reliance on technological tools to express emotions may reduce interactions with others, thus weakening social skills. While humans enjoy the convenience of emotional interactions with AI, they also realize that this dependence is not based on genuine emotional connections but is rather a technology-driven simulation. This paradox reflects both expectations for AI and anxieties about its potential threats. A14 expressed concerns about adolescents: "If teenagers are exposed to generative AI too early, it might make them think less comprehensively." Adolescents are at a stage where their cognitive and social abilities are still developing, and with limited knowledge of AI literacy, they may become overly reliant on generative AI, which could negatively affect their critical thinking and interpersonal skills.

From the loss of independent thinking to the degradation of information skills and the weakening of social abilities, the heavy reliance on generative AI poses a risk to users' core competencies. This phenomenon not only validates the perception of risks brought by technology but also intensifies users' fear of the potential negative consequences of AI.

3.2.2.3. Concerns about being replaced by AI: The uncertainty of career development. Dependence on generative AI also intensifies fears of career uncertainty, particularly as its efficiency and capabilities increasingly outpace human performance. The superior accuracy and productivity of generative AI in certain tasks have led some people to experience a heightened sense of career insecurity. A3 mentioned, "When writing code, I can truly feel that it outperforms me in this type of formatted work. It might really replace humans." This reflects the direct impact of technological advancements on traditional jobs, especially those that involve standardized and repetitive tasks. Similarly, A2 expressed concerns about the language translation industry: "Our professor said ChatGPT can now rival the level of many graduate students majoring in translation, and it really feels like we're going to be out of jobs."

Not only technical roles but also creative industries are facing challenges. A10 stated, "ChatGPT definitely poses a significant and potentially fatal threat to certain industries, like those in creative fields and public relations." The content generation capabilities of generative AI allow it to take on part of the creative work, causing professionals in these industries to worry about the diminishing value of their jobs. Additionally, A12 emphasized the potential threat AI poses across multiple fields: "If I need a piece of code, it can directly generate it for you. It's quite terrifying, and it could lead to mass unemployment." They also pointed out that AI-generated images might threaten designers' jobs, highlighting how generative AI is reshaping the professional competition landscape across different fields.

In response to the uncertainty brought about by generative AI, some participants have proposed possible ways to cope. A12 mentioned, "I think every industry needs to be alert to the possibility of being replaced by AI. What we can do now is to keep learning and improving." This perspective reflects a critical awareness of the threats posed by AI, emphasizing the importance of adaptability and innovation. However, this strategy may not completely alleviate anxiety for some, as the pace and depth of technological advancements could far outstrip an individual's ability to improve, resulting in ongoing pressure to learn and a sense of job insecurity.

In summary, from the replacement of standardized tasks to the challenges posed to cross-disciplinary creative work, and the increased pressure to learn, generative AI is profoundly impacting users' sense of job security.

3.2.3. Refining the factors influencing dependence on generative AI

Based on the qualitative analysis of the in-depth interviews, six factors were ultimately identified as potentially influencing dependence on generative AI (DGAI) and fear of AI (FAI): perceived usefulness (PU), perceived ease of use (PE), perceived risk (PR), perceived anthropomorphism (PA), self-efficacy (S), and AI literacy (AIL). The examples of the three-level coding process are shown in Table 1.

Therefore, based on the literature review and the results of the in-depth interviews, this study proposes the following research question: What is the relationship between the characteristics of generative AI (PU, PE, PR, PA) and user traits (E, S) with DGAI and FAI?

4. Study 2

The results from in-depth interviews summarized the factors influencing the use of generative AI and its impacts. Building on these insights and existing research, a survey was subsequently developed for examining the proposed research question.

Table 1. Example of three-level coding process.

Open coding	Axial coding	Selective coding
It can **quickly and efficiently** help me find answers or retrieve relevant information.	Perceived usefulness (PU)	Characteristics of GAI
It helps guide my reports and papers, provides more comprehensive analysis, and **assists me** in completing tasks and assignments more effectively.		
As a university student, I find it relatively **easy to get started**.	Perceived ease of use (PE)	
The **barrier to entry is low**, and anyone can use it if they want to.		
I don't dare to share personal information in it because I'm afraid of **data leaks**.	Perceived risk (PR)	
I worry that I might **lose the ability to think independently and creatively**.		
If I use it to write papers, I'm afraid it might **fail plagiarism checks**.		
It feels like an **older brother or sister** in my life—I turn to it when I face difficulties.	Perceived anthropomorphism (PA)	
When talking to it, I sometimes feel like I'm asking a **teacher** a question, so I unconsciously use **polite language**.		
It feels like **a person**—it even **gets angry**.		
I'm more willing to **step out of my comfort zone** when it comes to new technologies.	Self-efficacy (S)	Individual differences
Honestly, I'm **not particularly smart** and have to invest a lot of time and effort into my studies.		
I've started to **lose confidence** in my own abilities.		
I don't overestimate its abilities because I'm somewhat **aware of its limitations**.	AI literacy (AIL)	
I do have a bit of a **fascination with technology**.		
If one day this AI suddenly becomes unusable, I would feel very uncomfortable, as if there were **ants crawling on me**.	Dependence on generative AI (DGAI)	Dependence on generative AI (DGAI)
I feel that I really need it, and if it's gone, I would definitely **feel uneasy**.		
The widespread use of generative AI could exacerbate the **division of social and cultural classes** in our society.	Fear of AI (FAI)	Fear of AI (FAI)
When writing code, I can truly feel that it's stronger than me. In tasks like this, which are more structured, it could really **replace humans**.		

4.1. Method

Study 2 employed a two-stage survey process: a pilot survey followed by a formal survey. The pilot survey of 53 valid responses was conducted to evaluate the reliability and validity of the scales. Items that did not meet the required psychometric standards were removed. In addition, participants' feedback was used to refine ambiguous or difficult-to-understand wording in the questionnaire.

4.1.1. Participants and demographics

The formal survey was conducted later in March 2024 and distributed online via Chinese social media platforms such as WeChat and public academic accounts. Participation was voluntary and anonymous, with a small incentive offered to each respondent. The study obtained informed consent from all participants, and the data collected was used solely for academic research. After data cleaning, a total of 424 valid responses were retained for analysis.

All participants were Chinese nationals, with 260 females (61.32%) and 164 males (38.68%). In terms of age distribution, 57.78% were aged 23 to 27, and 30.42% were aged 18 to 22. Regarding education level, most participants held a bachelor's or master's degree (both 46.23%), indicating a relatively well-educated sample.

This questionnaire also investigated participants' experiences with generative AI technologies, including duration of use, frequency, daily usage time, and use cases. A total of 42.45% of participants reported using generative AI for between six months and one year, while 34.91% indicated they had been using it for over a year. On one hand, this suggests that most respondents have accumulated a certain level of experience with generative AI. On the other hand, approximately 77.36% (those who have used it for six months to one year and those over a year) have relatively long-term experience, indicating that generative AI may have already become integrated into their daily work, study, or life, potentially forming a foundation for habitual reliance—particularly among those who have used it for more than a year, as long-term usage may imply that the technology is gradually being incorporated into personal behavior patterns.

Secondly, data on usage frequency reveals potential differences in dependency levels. About 38.21% of participants reported using generative AI moderately, around three to six times per week, while 14.15% reported using it daily. From a dependency perspective, users with higher usage frequency may show a stronger tendency toward reliance on generative AI. This is especially true for daily users, who may increasingly view the technology as an indispensable part of their everyday routines.

In terms of daily usage time, most participants spent between 30 min to one hour (40.33%) or less than 30 min (34.67%) per day. However, 5.90% of respondents reported using it for more than two hours daily. These high-use individuals may exhibit stronger dependency, particularly in scenarios involving complex tasks or those requiring intensive engagement with generative AI.

Regarding usage purposes, participants reported employing generative AI in contexts such as learning, work, low-level tasks, entertainment, and emotional needs. Learning was the most common purpose, with 83.49% of respondents indicating this use case. This suggests that generative AI has become an indispensable tool in the learning process—especially for data processing, completing assignments, and skill development (e.g., language learning or programming). The high proportion of learning-related usage may be linked to cognitive dependence, wherein users rely on AI to improve efficiency, simplify complex tasks, and lower barriers to learning. Similarly, a large proportion of respondents (60.38%) reported using generative AI for work purposes, reflecting its widespread application in professional settings. This type of use may also be associated with psychological dependence—particularly when generative AI significantly enhances productivity or replaces traditional workflows, leading users to become more reliant on it. In contrast, entertainment (12.50%) and emotional support (11.79%) were less commonly cited usage scenarios, indicating relatively low levels of reliance on generative AI in these areas. Entertainment uses are typically optional and driven more by curiosity or novelty, hence associated with lower dependency. However, participants in this category may demonstrate a mild form of psychological reliance, such as deriving satisfaction or enjoyment from creative collaboration with AI. Emotional support was the least reported use case (11.79%), suggesting that most users have not yet viewed generative AI as a primary tool for addressing emotional needs or providing psychological support. This usage scenario might correspond to emotional dependence, but the low percentage suggests that users' emotional trust in generative AI remains limited.

Overall, the participant data provide both demographic context and behavioral evidence to assess the generalizability and depth of users' AI engagement. The findings reflect a sample with relatively high digital literacy and sustained exposure to generative AI technologies.

4.1.2. Design

The sources of the scales used to measure all the variables are as follows. Participants were asked to rate the following questions using a seven-point Likert scale (1 = "Strongly Disagree", 7 = "Strongly Agree"). All items and their sources are provided in Appendix 2.

PU, PE, and PR were measured through Technology Acceptance Model Edited to Assess ChatGPT Adoption (Sallam et al., 2023). The three dimensions comprise a total of 11 items, such as "I believe that using generative AI can save time and effort in my university assignments and duties", "Generative AI does not require extensive technical knowledge", and "I am concerned about the potential security risks of using generative AI". (PU's $\alpha = 0.834$, CR = 0.847, AVE = 0.496; PE's $\alpha = 0.761$, CR = 0.764, AVE = 0.619; PR's $\alpha = 0.660$, CR = 0.669, AVE = 0.401)

PA was measured using the adapted items from (Moussawi & Koufaris, 2019). The scale contains six items, with examples such as "Generative AI is able to speak like a human" and "Generative AI can be friendly". ($\alpha = 0.812$, CR = 0.775, AVE = 0.433)

S was measured through the General Self-Efficacy Short Scale–3 (Doll et al., 2021). The three items are: "I can rely on my own abilities in difficult situations", "I am able to solve most problems on my own", and "I can usually solve even challenging and complex tasks well". ($\alpha = 0.881$, CR = 0.881, AVE = 0.712)

AIL was measured with an adopted 12-item instrument from (Wang et al., 2023). Three items were dropped for better validity. Examples included "I can skillfully use generative AI to help me with my daily work" and "I always comply with ethical principles when using generative AI". ($\alpha = 0.860$, CR = 0.851, AVE = 0.411)

DGAI was adapted from the items from (Morales-García et al., 2024). It comprises five items, such as "I feel unprotected when I do not have access to generative AI" and "I constantly need validation or feedback from AI systems to feel confident in my decisions". ($\alpha = 0.815$, CR = 0.830, AVE = 0.493)

FAI was measured with the scale from (Sindermann et al., 2021). This scale comprises three items, namely "I fear AI", "AI will destroy humankind", and "AI will cause many job losses". ($\alpha = 0.658$, CR = 0.635, AVE = 0.435)

4.2. Results

According to the results of the mediation analysis (see Table 2), perceived risk exerted a significant direct positive effect on fear of AI ($\beta = 0.27$, *p* < 0.001), indicating that individuals with higher perceived risks of AI reported stronger fear. In contrast, the direct effects of other variables—perceived usefulness, self-efficacy, and perceived anthropomorphism—on FAI were nonsignificant (*p* > 0.05), suggesting their influence on fear might operate indirectly through mediators.

In terms of the mediating role of dependence on generative AI, perceived usefulness and perceived anthropomorphism significantly increased dependence on generative AI (PU: $\beta = 0.22$, *p* < 0.001; PA: $\beta = 0.15$, *p* = 0.006), while self-efficacy reduced dependence ($\beta = -0.13$, *p* = 0.004). Dependence on generative AI itself significantly intensified fear of AI ($\beta = 0.173$, *p* < 0.001), implying that greater reliance on AI tools correlates with heightened anxiety. Combined with earlier mediation tests (bootstrap confidence intervals), PU, S, and PA exhibited significant indirect effects via DGAI, whereas PR and PE did not.

The moderated mediation analyses revealed three key findings (see Table 3). Perceived usefulness and perceived anthropomorphism exhibited significant conditional indirect effects on fear of AI through dependence on generative AI, moderated by AI literacy. For perceived usefulness, the indirect

Table 2. Standardized and unstandardized estimates for mediation analysis.

Path	β (std.all)	Unstd. (estimate)	SE	95% CI	z	p
Direct effects (FAI ∼)						
PR → FAI (c1)	.27**	0.230**	0.040	[0.15, 0.31]	5.75	<.001
PU → FAI (c2)	−.08	−0.085	0.058	[−0.20, 0.03]	−1.47	.142
PE → FAI (c3)	−.05	−0.044	0.046	[−0.14, 0.05]	−0.96	.335
S → FAI (c4)	−.06	−0.060	0.060	[−0.18, 0.06]	−1.00	.317
PA → FAI (c5)	.02	0.019	0.046	[−0.07, 0.11]	0.40	.687
Mediator paths (DGAI ∼)						
PR → DGAI (a1)	.10	0.097	0.054	[−0.01, 0.20]	1.78	.074
PU → DGAI (a2)	.22***	0.279***	0.073	[0.14, 0.43]	3.84	<.001
PE → DGAI (a3)	−.06	−0.058	0.053	[−0.16, 0.04]	−1.11	.269
S → DGAI (a4)	−.13**	−0.167**	0.058	[−0.28, −0.05]	−2.90	.004
PA → DGAI (a5)	.15**	0.167**	0.061	[0.05, 0.29]	2.72	.006
Outcome path						
DGAI → FAI (b)	.20***	0.173***	0.043	[0.09, 0.26]	4.02	<.001

Note: N = 424. Unstd. = unstandardized estimate; β = standardized estimate; CI = confidence interval.
**p* < .05.
***p* < .01.
****p* < .001.

Table 3. Conditional indirect effects of predictors on fear of AI through dependence on generative AI, moderated by AI literacy.

Predictor	Conditional indirect effect	β (std.all)	95% CI	p	Interaction (DGAI × AIL) β	95% CI	p
PU	Low AIL (-1SD)	0.243*	[0.036, 0.502]	.044	−0.455	[−0.185, 0.014]	.096
	High AIL (+1SD)	0.028*	[0.044, 0.376]	.028			
PR	Low AIL (-1SD)	0.137	[0.006, 0.243]	.123	−0.421	[−0.168, 0.016]	.121
	High AIL (+1SD)	0.016	[0.008, 0.182]	.099			
PA	Low AIL (-1SD)	0.234	[0.027, 0.431]	.066	−0.451	[−0.181, 0.014]	.097
	High AIL (+1SD)	0.025*	[0.030, 0.320]	.046			
S	Low AIL (-1SD)	−0.132	[−0.319, −0.008]	.126	−0.453	[−0.180, 0.014]	.092
	High AIL (+1SD)	−0.016	[−0.239, −0.011]	.103			
PE	Low AIL (-1SD)	0.026	[−0.067, 0.114]	.659	−0.455	[−0.184, 0.013]	.095
	High AIL (+1SD)	0.003	[−0.051, 0.086]	.653			

Note: N = 424. β = standardized estimates; CI = bias-corrected bootstrap confidence intervals (5,000 samples).
**p* < .05. Non-significant paths (p ≥ .10) omitted for clarity.

effect was stronger at low AIL ($\beta = 0.243$, *p* = .044, 95% CI [0.036, 0.502]) than at high AIL ($\beta = 0.028$, *p* = .028), with a marginally significant interaction ($\beta = -0.455$, *p* = .096). For perceived anthropomorphism, high AIL attenuated the indirect effect ($\beta = 0.234$ vs. 0.025, *p* = .046), though the interaction term approached significance ($\beta = -0.451$, *p* = .097).

Self-efficacy showed a trend where higher AIL reduced the negative indirect effect ($\beta = -0.132$ vs. -0.016), but neither the interaction nor indirect effects reached significance (*p* > .10). Perceived risk and perceived ease of use demonstrated no statistically significant moderated mediation patterns (*p* > .10).

4.3. Discussion

The current study elucidates the dual pathways through which cognitive and affective appraisals shape fear of AI, particularly highlighting the distinct roles of direct risk perception and technology-dependent mediation. The findings offer three pivotal contributions to the literature on attitudes towards AI.

4.3.1. The direct effect of perceived risk

Perceived risk refers to an individual's anticipation of potential issues that may arise from the introduction of a technology or method (Lee, 2009). Featherman and Pavlou (2003) defines perceived risk as the potential losses that may occur while pursuing desired outcomes. The robust direct effect of perceived risk on fear of AI ($\beta = 0.27$, *p* < 0.001) underscores the primacy of threat appraisal in driving AI-related fear. Unlike other predictors, the influence of perceived risk operates independently of dependence on generative AI, suggesting that intuitive risk perceptions—such as concerns about job displacement or ethical breaches—may bypass rational evaluations of AI utility or competence. This aligns with risk-as-feelings theory (Loewenstein et al., 2001), where visceral fear responses dominate over deliberative cognitive processes when risks feel immediate or uncontrollable. Practically, this implies that mitigating AI fear requires directly addressing existential and societal risks rather than solely focusing on usage patterns.

4.3.2. The mediating role of dependence on generative AI

The mediating role of dependence on generative AI reveals a paradoxical dynamic: while *perceived usefulness* and *perceived anthropomorphism* enhance AI adoption (PU: $\beta = 0.22$; PA: $\beta = 0.15$), they simultaneously amplify fear through increased dependence (DGAI $\rightarrow$ FAI: $\beta = 0.17$). This duality mirrors the "double-edged sword" phenomenon observed in other technologies (e.g., social media), where functional benefits coexist with psychological costs (Zhang & Mao, 2025). For instance, users who rely heavily on AI for productivity (high PU) or perceive AI as human-like (high PA) may develop anxiety about over-reliance or loss of autonomy. Conversely, *self-efficacy* acts as a protective factor, reducing dependence ($\beta = -0.13$) and thereby indirectly alleviating fear. This supports *protection motivation theory* (Rogers, 1975), where confidence in one's ability to manage threats (here, AI dependence) diminishes maladaptive responses.

4.3.3. The absence of mediation for perceived ease of use

The absence of significant mediation for perceived ease of use challenges conventional assumptions in technology adoption models (e.g., TAM (Davis 1989)). While ease of use typically drives adoption, its dissociation from fear suggests that effortless AI interactions may not inherently provoke anxiety unless coupled with high utility or human-like attributes. This invites a reevaluation of AI design principles: minimizing friction in user experience may enhance adoption without exacerbating fear, provided utilitarian and anthropomorphic features are carefully balanced.

4.3.4. The moderating role of AIL

The moderated mediation analyses further clarify how AI literacy shapes the psychological consequences of generative AI adoption, revealing nuanced boundary conditions for the observed mediation pathways.

First, the buffering role of AI literacy in mitigating fear driven by perceived usefulness and perceived anthropomorphism aligns with the technology empowerment framework (Hoffman & Novak, 2018). For individuals with high perceived usefulness, dependence on AI tools amplified fear significantly when AI literacy was low ($\beta = 0.243$) but diminished to negligible levels when AI literacy was high ($\beta = 0.028$). This pattern suggests that technical competence disrupts the "efficiency-anxiety paradox" discussed above—where reliance on highly useful AI breeds fear of overdependence. Similarly, high AI literacy attenuated the fear-enhancing effect of anthropomorphism (PA: $\beta = 0.234$ to 0.025), likely because literate users better distinguish human-like AI behaviors from actual human agency, reducing uncanny valley effects (Mori et al., 2012). Though the interactions were marginally significant (*p* ≈ .096–.097), their effect sizes ($\beta \approx -0.45$) imply practical relevance, echoing findings that even subtle literacy gains can foster resilience against AI anxiety (Schiavo et al., 2024).

Second, the non-significant moderation for self-efficacy and perceived risk highlights divergent mechanisms in fear regulation. While self-efficacy directly reduced dependence (as shown earlier), its interplay with AI literacy did not amplify this protective effect. This may reflect ceiling effects: high self-efficacious individuals might already possess coping strategies that render additional literacy redundant. Conversely, the direct fear-driving mechanism of perceived risk appears resistant to the moderating influence of AI literacy, reinforcing the notion that existential risks operate independently of technical knowledge.

Third, the null results for perceived ease of use also challenge assumptions in user experience design. While ease of use typically fosters adoption, its dissociation from moderated fear pathways suggests that intuitive interfaces alone cannot alleviate anxiety rooted in utility (PU) or human-like design (PA). This underscores the need for literacy-integrated design—pairing user-friendly AI with embedded educational cues (e.g., transparency about AI limitations)—to preempt dependence-driven fear.

5. General discussion

Overall, this study collectively advances the cognitive-affective model of AI fear by delineating how rational appraisals (perceived usefulness, perceived anthropomorphism) and emotional risk perceptions (perceived risks) operate through divergent pathways. They also underscore the critical role of self-efficacy as a mitigator of dependence-driven fear, which offers actionable insights for interventions.
The moderation results also extend the cognitive-affective model of technology adaptation by positioning AI literacy as a critical moderator that decouples functional dependence from maladaptive fear.

5.1. Beyond definitions: Deepening the role of AI literacy in reducing fear and promoting adoption

This study provides empirical evidence demonstrating the critical role of AI literacy in mitigating fear of AI arising from users' dependence on generative AI. The findings suggest that users' patterns of dependence on generative AI may be more context-dependent and adaptive than often portrayed, and that excessive fear of relying on AI may be unwarranted.

In extending Zhan et al. (2024)'s framework, this study offers a more nuanced understanding of how dependence on generative AI shapes users' emotional responses. While prior research has suggested that the perceived affordances of AI technologies can help alleviate fear, this study offers a more nuanced view by showing how users' actual dependence on generative AI may shape their emotional responses. Specifically, the results contribute to a deeper understanding of the cognitive, emotional, and behavioral mechanisms that underlie individuals' perceptions of emerging technologies. These insights help clarify what factors may hinder or facilitate the acceptance of AI.

While this study emphasizes the importance of AI literacy in alleviating fear and promoting more confident use of generative AI, it acknowledges the concern that such a claim may seem tautological if not further elaborated. Therefore, this paper aims to unpack how AI literacy contributes to users' cognitive and emotional responses to AI, and what actionable steps can be taken to foster meaningful literacy. Existing literature suggests that AI literacy should go beyond mere awareness of AI's capabilities and limitations. It involves the development of critical understanding about how AI works, where it may go wrong, and how users can evaluate, challenge, and appropriately use AI outputs (Ng et al., 2021). This study observed that users with higher AI literacy levels tended to report less fear, more adaptive coping strategies (e.g., cross-verifying AI outputs). This aligns with findings from studies on digital resilience, which suggest that knowledge and transparency can improve users' ability to respond confidently to digital challenges and safely navigate the digital world (Lee & Hancock, 2023).

Moreover, this study highlights that AI literacy must extend beyond raising general awareness of AI's capabilities and limitations. A meaningful literacy framework should equip users to understand how AI systems function, recognize potential risks such as bias or misinformation, and critically engage with AI-generated content (Ng et al., 2021). Currently, the AI industry focuses on developing more engaging and entertaining products, with little attention paid to the long-term benefits of improving users' AI literacy for shaping positive attitudes toward AI. In terms of practice, this study's results support the development of user-centered and ethically grounded AI literacy initiatives. These should not only explain the "what" of AI, but also offer tools for navigating ethical dilemmas, privacy concerns, and disinformation risks. For example, adopting realism-inspired frameworks such as the one proposed by can help integrate transparency, documentation, and accountability into both educational materials and AI tool design. This study calls on the AI industry to take social responsibility and prioritize enhancing AI literacy. By raising the overall AI literacy levels among users, misunderstandings and improper usage can be mitigated, fostering human-AI collaboration and societal acceptance while promoting the sustainable development of AI across various domains.

In conclusion, this study argues that promoting AI literacy must be part of a broader ecosystem that includes concrete, actionable ethical guidelines. Future research should investigate how combined interventions—enhancing AI literacy while embedding ethical safeguards—can sustainably improve users' trust, critical thinking, and long-term engagement with AI technologies (Lin, 2024).

5.2. Limitations and future research

Despite efforts to carefully adapt established scales, some overlaps between constructs were observed. For example, certain perceived risk items may also reflect concerns about dependence on AI (e.g., "I think that relying on generative AI can disrupt my critical thinking skills"), and some AI literacy items could conceptually overlap with perceived ease of use, perceived usefulness, or perceived risk. While these overlaps do not necessarily compromise the overall validity of the constructs, they may reduce discriminant validity and should be addressed more systematically in future research.

Additionally, this study employed a general self-efficacy scale rather than an AI-specific self-efficacy measure. Although this choice was guided by practical considerations during survey development, future work should consider using domain-specific measures, such as the GSE-6AI scale (Morales-García et al., 2024), to more precisely capture self-efficacy in AI-related contexts. Addressing these measurement refinements would help clarify the relationships between constructs and strengthen the robustness of findings.

This study confirmed the relationship between dependence on generative AI and fear of AI. More studies could further investigate how dependence on generative AI influences individuals' fear of AI and explore the mechanisms and influencing factors underlying this relationship. Such investigations would contribute to a more comprehensive understanding of the attitude formation process toward AI.

One of the limitations of this study lies in the sample composition. All participants were recruited from China, and the majority were young adults, primarily in their early twenties. This demographic specificity may influence the findings in several ways. First, China's strong governmental support for AI development has contributed to a relatively positive public attitude toward generative AI compared to other countries (Sindermann et al., 2022)

This study offers a cross-sectional snapshot of users' experiences and self-reported perceptions of dependence on generative AI. However, it does not provide direct evidence of the potential cognitive, behavioral, or social consequences of such reliance. While the survey and interviews revealed a range of user attitudes—ranging from enthusiasm to concern—these should not be interpreted as causal outcomes. Future research could benefit from experimental or longitudinal designs to examine how different levels or patterns of generative AI use might affect users' critical thinking, creativity, interpersonal skills, or emotional well-being over time. For instance, longitudinal tracking could reveal whether prolonged use of or even reliance on generative AI affects users' ability to solve open-ended problems or engage in collaborative communication. Incorporating such methods would help clarify not only the existence of dependence, but also its potential benefits or costs, offering a more comprehensive understanding of the human-AI relationship.

6. Conclusion

AI literacy serves as a foundational measure for understanding AI accurately, using it appropriately, adapting to the changes it brings, and enjoying the benefits it offers. This study explores the complex relationship between users' dependence on generative AI and their fear of AI under varying levels of AI literacy. The findings reveal that users develop dependence on generative AI due to their varying needs. The progression of this dependence can be divided into three stages: functionality exploration and tool acceptance, functionality deepening and habit formation, and psychological dependency and multi-scenario application. This dependence can trigger various concerns, including concerns about privacy, personal capabilities, and career development. Through the mediation and moderation analysis, this paper found that perceived risk had a significant direct positive effect on fear of AI, while perceived usefulness, self-efficacy, and perceived anthropomorphism exerted indirect effects through dependence on generative AI, with perceived usefulness and perceived anthropomorphism increasing dependence and self-efficacy reducing it. Also, AI literacy significantly buffered the indirect effects of perceived usefulness and perceived anthropomorphism on ear of AI via dependence on generative AI (effects were stronger at low AI literacy levels), but showed no significant moderating role for perceived risk, perceived ease of use or self-efficacy pathways, suggesting technical competence primarily mitigates anxiety risks from utilitarian and anthropomorphic dependence. Overall, this study clarifies the multifaceted factors influencing users' attitudes toward generative AI and the different patterns observed across varying levels of AI literacy. It provides theoretical understanding of the psychological mechanisms underlying users' attitudes toward generative AI and offers practical implications for designing interventions. In the promotion of AI technologies, more attention should be paid to enhancing users' cognitive understanding of AI, as this approach can reduce the fear of (relying on) AI and encourage a more positive and open attitude toward the development of AI.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT to improve writing quality. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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About the author

Jiahui Liu is a PhD student in the Department of Media and Information at Michigan State University. Her research focuses on human-AI interaction and media psychology. She is particularly interested in exploring why, how, and what happens when humans and AI communicate emotionally.

Appendix 1 Interview outline

1. Users' needs and purposes for using generative AI

1. What do you most frequently use generative AI for?
2. In what situations are you more inclined to use generative AI?
3. What specific functionalities or experiences does generative AI provide for you?
4. Do you find using generative AI helpful? If No, why not? If Yes, in what aspects?
5. What are your expectations for generative AI?

2. Users' evaluation of generative AI

1. Overall, how do you feel about using generative AI?
2. Do you feel dependent on generative AI? Why or why not?
3. What potential risks do you think generative AI may bring? How do you evaluate the convenience and risks associated with ChatGPT?

Appendix 2

Table A1. Variables and items in the questionnaire.

Variables		Items	References
Perceived usefulness	PU1	Generative AI helps me save time when searching for information.	Sallam et al., 2023
	PU2	For me, generative AI is a reliable source of accurate information.	
	PU3	I recommend generative AI to my classmates, colleagues, and friends to facilitate their tasks.	
	PU4	Generative AI is more useful than other sources of information that I have used previously	
	PU5	I appreciate the accuracy and reliability of the information provided by generative AI.	
	PU6	I believe that using generative AI can save time and effort in my study and work.	
Perceived ease of use	PE1	It does not take a long time to learn how to use generative AI.	
	PE2	Using generative AI does not require extensive technical knowledge.	
Perceived risk	PR1	I am concerned that using generative AI would get me accused of plagiarism.	
	PR2	I am concerned about the potential security risks of using generative AI.	
	PR3	I think that relying on generative AI can disrupt my critical thinking skills.	
Perceived anthropomorphism	PA1	Generative AI is able to speak like a human.	Moussawi & Koufaris, 2019
	PA2	Generative AI can be happy.	
	PA3	Generative AI can be friendly.	
	PA4	Generative AI can be respectful.	
	PA5	Generative AI can be funny.	
	PA6	Generative AI can be caring.	
Self-efficacy	S1	I can rely on my own abilities in difficult situations.	Doll et al., 2021
	S2	I am able to solve most problems on my own.	
	S3	I can usually solve even challenging and complex tasks well.	
AI literacy	AIL1	I can distinguish between smart devices and non-smart devices.	Wang et al., 2023
	AIL2	I know how AI technology can help me.	
	AIL3	I can identify the AI technology employed in the applications and products I use.	
	AIL4	I can skilfully use AI applications or products to help me with my daily work.	
	AIL6	I can use AI applications or products to improve my work efficiency.	
	AIL7	I can evaluate the capabilities and limitations of an AI application or product after using it for a while.	
	AIL8	I can choose a proper solution from various solutions provided by a smart agent.	
	AIL9	I can choose the most appropriate AI application or product from a variety for a particular task.	
	AIL10	I always comply with ethical principles when using AI applications or products.	

(*continued*)

Table A1. Continued.

Variables		Items	References
Dependence on generative AI	DGAI1	I feel unprotected when I do not have access to generative AI.	Morales-García et al, 2024
	DGAI2	I'm concerned about the idea of being left behind in my tasks or projects if I do not use generative AI.	
	DGAI3	I do everything possible to stay updated with generative AI to impress or remain relevant in my field.	
	DGAI4	I constantly need validation or feedback from generative AI systems to feel confident in my decisions.	
	DGAI5	I fear that generative AI might replace my current skills or abilities.	
Fear of AI	FAI1	I fear artificial intelligence.	Sindermann et al., 2021
	FAI2	Artificial intelligence will destroy humankind.	
	FAI3	Artificial intelligence will cause many job losses.	